



Diamond Energy's East Kalimantan Mine Acquisition: Investment Analysis

Financial appraisal covering free cash flows, WACC, NPV, IRR, payback, profitability index, capital structure effects, and risks, to provide an investment recommendation.

Executive summary

Proposal

Acquire mine (12M tons reserves, ~1M tons/year production). CapEx: \$30M (\$15M in 2015, \$15M in 2016). Working capital: \$2.5M/year (from 2015).

Key financial inputs

Beta: 1.4; Risk-free: 7.5%; ERP: 8.8%; Debt: \$12M @14%; Tax: 25%. Equity covers remaining \$18M + WC.

Decision focus

Compute WACC & FCF. Evaluate NPV, IRR, payback, PI. Assess capital structure sensitivity and scenario risk.



Free Cash Flow (FCF) Identification

- Project FCF is derived from Exhibit 9 figures, using yearly data from 2015-2026. Taxes are applied at 25% to pre-tax operating profit.
- $FCF = \text{After-tax operating profit} + \text{Depreciation} - \text{CapEx} - \Delta \text{Working Capital}$
- Depreciation = \$3,000k/year (\$30M/10 years)
- CapEx and WC cash outflows are detailed in Exhibit 9.

WACC Calculation (Base Case)

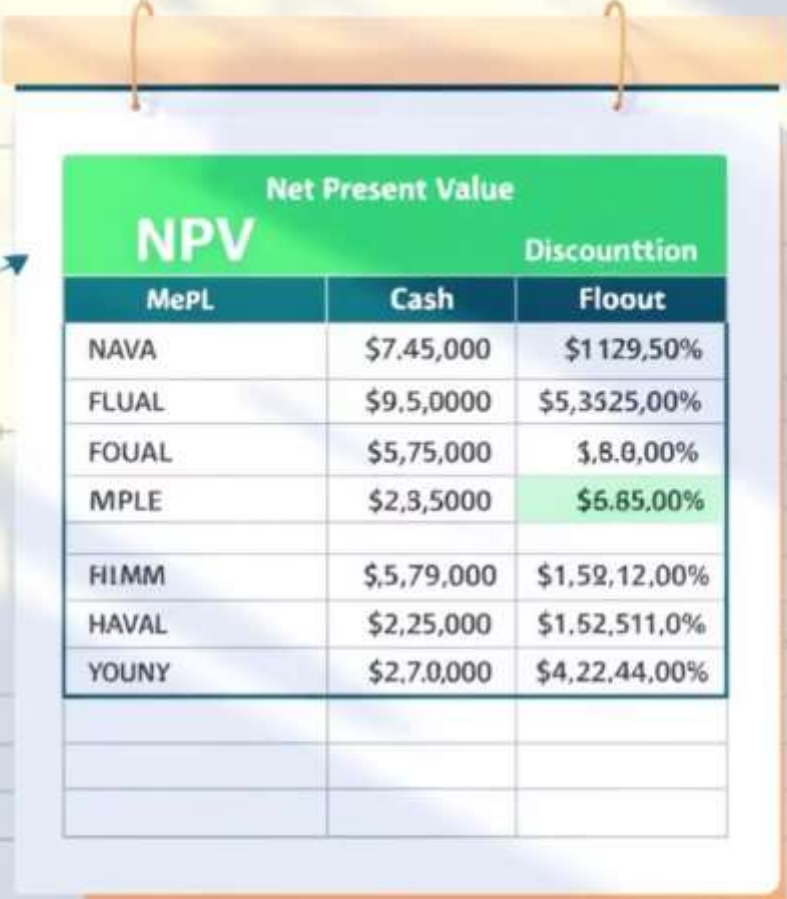
Key inputs: $r_D = 14\%$, $D = \$12\text{M}$, $E = \$18\text{M}$, Tax rate = 25%.

1. $r_E = 7.5\% + 1.4 \times (8.8\%) = 19.82\%$
2. Capital weights: $D/(D+E) = 0.40$; $E/(D+E) = 0.60$
3. After-tax cost of debt = $14\% \times (1 - 0.25) = 10.5\%$

$\text{WACC} = 0.40 \times 10.5\% + 0.60 \times 19.82\% = 16.09\%$
(rounded)



NPV, IRR, Payback & Profitability Index (summary)



Net Present Value		
NPV		Discounttion
MePL	Cash	Floout
NAVA	\$7,45,000	\$1129,50%
FLUAL	\$9,5,0000	\$5,3525,00%
FOUAL	\$5,75,000	\$,8.0,00%
MPLE	\$2,3,5000	\$6.85,00%
FIIMM	\$,5,79,000	\$1,52,12,00%
HAVAL	\$2,25,000	\$1,52,511,0%
YOUNY	\$2,7,0,000	\$4,22,44,00%

Discounting FCFs (2015–2026) with WACC = 16.09% yields:

- NPV: ~\$1.2M (modest positive)
- IRR: $\approx 17.6\%$ (slightly above WACC)
- Payback: ~ 6.0 years (undiscounted)

Profitability Index: ≈ 1.04 (>1)

Interpretation: Base-case returns marginally exceed cost of capital; project value creation is sensitive to assumptions.

Capital structure impact on WACC (sensitivity)



Increased Debt (D/E ratio)

Targeting higher debt (D/(D+E)=0.60) could lower WACC to ~14.23% if rE doesn't dramatically increase.



Risk of Higher Leverage Costs

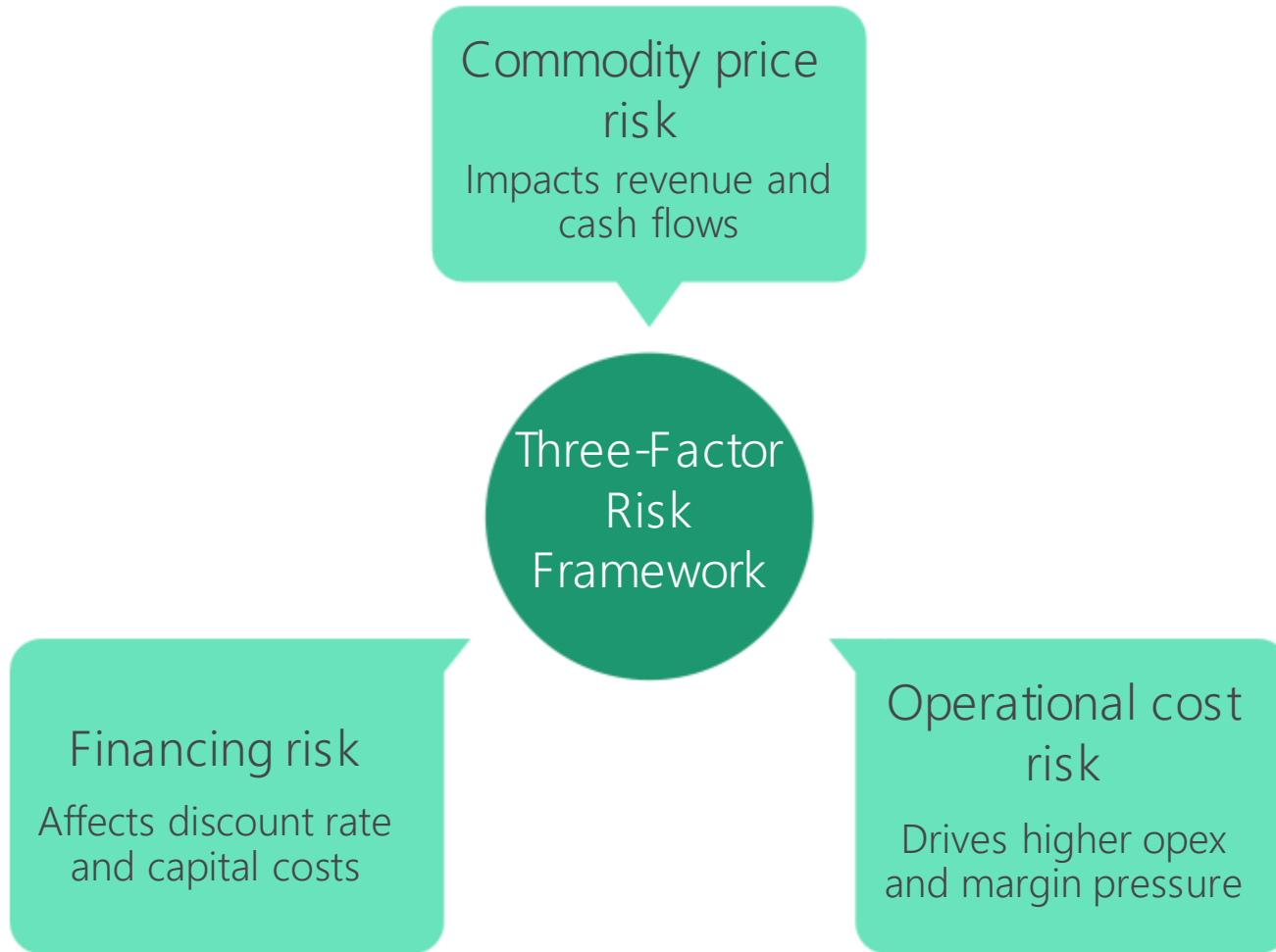
Higher debt increases financial risk, potentially raising rE and rD. Industry volatility and existing leverage caution against aggressive debt increases.



Recommendation

Modest debt increase can lower WACC and improve NPV. Stress test rE/rD under distress scenarios for sensitivity analysis.

Risk & uncertainty analysis



Major risk drivers:

- **Coal price volatility:** Downside (price below \$45/ton) compresses margins.
- **Operational cost inflation:** Opex grows 5% p.a.; exchange rates/logistics add risk.
- **Financing & policy risk:** 14% debt cost and government limits/taxes create cash flow uncertainty.

Recommendation: Conduct scenario analysis (base, downside -25% price, upside +20%) to recompute NPV/IRR/debt service. Project is highly price-sensitive.

Scenario outcomes (illustrative)

Base case

WACC 16.09%, NPV $\approx$ +\$1.2M, IRR $\approx$ 17.6%, PI $\approx$ 1.04. Marginally acceptable if strategic benefits exist.

Downside (price -25%)

NPV $\approx$ -\$8M to -\$12M. IRR $<$ WACC; payback exceeds project life. High probability given recent price environment.

Upside (price +20%)

NPV $>$ \$8M, IRR $>$ WACC, quicker payback. Depends on sustained market recovery and export demand.



Financing & bankability considerations



Bankers scrutinize debt service coverage, collateral, and parent guarantees. A \$12M loan at 14% interest demands robust project cash flow coverage under stress.

- Recommended mitigants: staggered drawdown, reserve-based covenants, partial price hedging, parent equity cushion.
- Credit view: Diamond's improving leverage helps, but sector sector volatility and scale limit lender appetite without strong guarantees.

Recommendation & next steps

Conditional approval. Project has marginal positive NPV but is highly sensitive to coal price and financing terms. Approve only if parent provides an equity guarantee, bank offers tighter covenants/better terms, and management implements hedging/cost controls.

Immediate actions

- Refine cash-flow model (monthly, currency).
- Conduct downside stress tests and covenant breach simulations.
- Negotiate debt tenor, grace, and partial price hedges.

Go/no-go criteria

- Approve if bank terms reduce WACC below IRR sensitivity, or parent guarantee improves financing.
- Decline if downside NPV remains negative or equity dilution exceeds limits.